Predicting Post-Earnings Announcement Drift Using NLP

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Abstract

We investigate whether earnings call transcripts contain predictive signals for short-term stock price direction, focusing on the 1-5 day window after announcements. Using a cleaned dataset of full transcripts, we fine-tune FinBERT and experiment with LLaMA-3.1 via instruction-based prompting to model executive tone and implied sentiment. While most individual models struggle to beat the majority-class baseline, an ensemble approach – predicting "Up" only when both models agree – shows a notable improvement in accuracy. These findings suggest that combining domain-specific encoding with decoder-style generation could help surface subtle directional signals, particularly in the context of long-only strategies.

1 Introduction

Predicting short-term stock price movement after earnings announcements has long been a central question in finance. A well-known phenomenon called post-earnings announcement drift (PEAD) persists even after efficient-market-style pricing mechanisms act, where earnings disclosures often spark large price jumps, but a significant drift in the direction of the surprise still unfolds over the following days or weeks (Christensen et al., 2025). Explaining why this happens – and whether unstructured text like earnings call transcripts contains predictive signals beyond the numeric data – remains an open challenge.

Most prior work has focused on structured data, aggregate sentiment from transcripts, or models trained to forecast long-term price drift, often well beyond one month post announcement (Medya et al., 2022; Chung et al., 2023; Zhu et al., 2025). While valuable, such designs do not isolate the contribution of managerial language itself, especially over short horizons. We hypothesize that executives communicate subtle cues – tone, emphasis, hesitations – that may be especially relevant in the 1-5 trading-day window following a call.

This paper pursues two primary contributions. First, we design a treatment vs. control experiment to isolate language-derived predictive power. The treatment model takes only transcript text as input, while the control is a majority-class/structured data baseline. Second, we explore domain-adapted approaches – fine-tuning FinBERT and experimenting with LLMs via prompting and confidence encoding – to handle long earnings transcripts and weak signals. We find that a combined strategy, where both models agree on the "Up" class, generates an accuracy edge (~54.60%) over the baseline (~53.67%). Given the difficulty of downward predictions, we propose a long-only tactic, leveraging upward predictions only when both models agree.

2 Background

2.1 Earnings-Call Text & Drift

Medya et al. (2022) studied ~100,000 earnings calls and found that semantic features extracted from transcripts outperformed traditional accounting metrics or analyst recommendations at predicting post-call stock moves. Chung et al. (2023) extended this by incorporating both textual and contextual inputs, achieving better PEAD forecasts than structured data alone. Zhu et al. (2025) further developed a multitask neural approach that includes investor-response sentiment as auxiliary tasks, significantly improving predictive performance over numeric methods.

2.2 Human vs. AI in Call Content

Bai et al. (2023) introduced an innovative information measure called HAID – the semantic gap between executive answers and ChatGPT's predicted answers to the same Q&A prompts – and found that higher HAID correlates with greater post-call volatility, trading volume increases, and delayed analyst forecast updates. This suggests that deviations from "AI-expected" language encode real investor-relevant signals.

2.3 Financial Language Models

On the modeling side, FinBERT is a BERT variant pretrained on large financial corpora that outperformed general-purpose BERT by ~15% on financial classification tasks (Araci, 2019). More recent architectures illustrate the growing specialization trend:

- BloombergGPT (Wu et al., 2023) is a closed-source 50B-parameter model trained on 363B financial tokens that is primarily for research use.
- FinGPT (Yang et al., 2023) is open-source and uses LoRA-based adaptation, but deploying it involves setting up substantial data pipelines and fine-tuning infrastructure beyond our resource and time constraints.
- FinBERT2 (Xu et al., 2025) achieves up to 3.3% accuracy gains over the original FinBERT across classification tasks, but at the time of this work it was not readily accessible and lacked integration with drift-prediction workflows.

Consequently, we chose to start with the established FinBERT model, which offered a strong yet accessible foundation for exploring language-based drift prediction.

3 Methods

3.1 Problem Formulation & Experimental Framework

We frame the task as binary classification: predict whether the sign of the stock return five trading days after an earnings call is positive (Up = 1) or non-positive (Down = 0) (we call this binary variable Close_5_dir). Since ~53.67% of calls result in Up, a majority-class baseline already scores ~53.67%. We compare this control with our treatment groups of text-only models trained only on the transcripts.

Other horizons (same-day exit, +1-day, +2-days) were also computed, but the 5-day window is treated as core (results for alternatives in Appendix A.3). Furthermore, we report overall accuracy, as well as class-specific metrics (precision, recall, F1) for the Up class – since only up-predictions are actionable under a long-only deployment.

3.2 Data Preprocessing

We collected ~20,000 S&P 500 earnings call transcripts (2015-2024) from the Wharton Research Data Services (WRDS), a web-based data service from the Wharton School of the University of Pennsylvania that provides access to a vast collection of financial, economics, and public policy databases. Only the prepared remarks and the main presenter speech sections were included in the analysis, as we aim to capture management tone and messaging uniformly, and Q&A often includes narrative drift unrelated to forward guidance and varies by analyst quality. Stock price returns are retrieved via yfinance, with Close_5_dir computed using closing price after 5 days vs. the opening price on day 0.

The data was randomly shuffled across all years and sectors to avoid temporal or sector-specific bias. It was then split 80/20 into training and validation sets, mimicking a cross-sectional deployment split rather than a time-series holdout. A random split maximizes representativeness and helps error analysis across sectors without leaking events.

3.3 Modeling Approaches

3.3.1 Baseline: Majority-class Predictor

Our best baseline – which serves as a control for text-derived lift – is a simple majority-class predictor which always predicts the "Up" class. Within our evaluation window, this approach achieves an accuracy of roughly 53.67%.

3.3.2 ModernBERT (Encoder-Only)

We first tested a general-purpose BERT-based model with the full transcripts, fine-tuned for drift prediction. However, only the classifier head was trained due to compute constraints. The performance did not exceed baseline, confirming the need for a domain-specific model.

3.3.3 FinBERT Fine-Tuning

FinBERT, pretrained on financial text and fine-tuned on sentiment benchmarks, provides up to ~15 percentage points improvement over generic BERT for financial sentiment tasks (Araci, 2019). We adapt FinBERT for binary drift classification by replacing its sentiment head with a custom 2-class softmax layer. Because of the 512-token limit, we fed the first chunk only, which generally contains executive opening remarks. This choice is motivated by executive "tone-setting" often appearing early, which may correlate with sentiment and drift. We tested three variants:

- Test 1: Train only the classifier layer
- Test 2: Train classifier + pooled output layer
- Test 3: Train classifier, pooler, and final transformer layers (9-12)

3.3.4 Chunked FinBERT + BiLSTM

To address token-limit issues, transcripts are divided into sixteen 512-token chunks (padding/truncating to 8192 tokens total), which is able to accommodate over 99% of our transcripts. Each chunk is encoded via FinBERT, and the CLS embedding from each chunk feeds into a bidirectional LSTM to model sequential tone shifts across a call. FinBERT layers are frozen to reduce compute cost and avoid overfitting, while the LSTM + classifier layers are trainable.

3.3.5 Sentiment Audit (FinBERT without Fine-Tuning)

Using pretrained FinBERT's original 3-class sentiment head (positive, neutral, negative), we run chunk-level inference across calls and average the softmax logits to produce a transcript-level sentiment distribution. This audit tests our hypothesis that earnings calls are structurally biased toward optimism, potentially explaining model misbehavior.

3.3.6 LLaMA-3.1 Instruction-Based Decoder Model with Confidence Labels

Meta's LLaMA-3.1 is a decoder-only LLM with up to a 128k token context window and open-weight availability, with instruction-tuned performance that rivals GPT-4 on reasoning benchmarks like MMLU (Llama Team, 2024). This made it ideal for tone-aware inference across long documents. We used a text-generation pipeline under quantized mode with a system + user prompt to force 2-word responses: *confident up*, *unconfident up*, *unconfident down*, or *confident down*. Prompt engineering emphasizes brevity, context-awareness, and tone sensitivity. Each call is queried **three times** independently (see Appendix A.1), and the counts of each label across runs ultimately become features for a classification model trained to predict drift (using logistic regression or random forest). This design captures not only the predicted direction but also model "confidence consistency" (i.e., whether the model repeatedly returns the same label over multiple runs), which we hypothesize corresponds to clearer tonal cues.

3.3.7 Ensemble Strategy

After observing the FinBERT (positive-subset) and LLaMA (Up) both individually tilt heavily toward Up recognition, we test the hypothesis that agreement between the models' predictions on Up yields improved accuracy. For each call, if both models predict Up, we label Up; otherwise, we abstain or default to baseline. This ensemble reflects the long-only strategy supported by data and practical retail constraints. We do this because 1) short selling is constrained for most retail investors (margin requirement, recall risk) (Kelly et al., 2016) and such, 2) down-class is under-represented with high-variance, and 3) even small precision gains on Up can be meaningful.

3.4 Evaluation Metrics & Diagnostics

We report accuracy, and – where available – precision, recall, F1 for the Up class (since Up is most relevant to long-only traders). Error analysis includes confusion between classes, skew toward Up predictions across all models, behavior across time horizons (1, 2, 5 trading days), and tone-based model failure modes (e.g. opening bias). Lastly, we avoid claims about profitability, instead noting relative accuracy lift as evidence for signal.

4 Results & Discussion

4.1 Summary of Main Results

Model	Overall Accuracy	Up-Precision	Up-Recall	Up-F1
Majority baseline (Up always)	53.67%	N/A	N/A	N/A
ModernBERT	~52.50%	–	–	–
FinBERT test 1 (classifier only)	53.67%	53.67%	100%	69.85%
FinBERT test 2 (pooler + clf)	52.74%	53.42%	93.19%	67.91%

FinBERT test 3 (above plus transformer layers 9-12)	51.99%	53.82%	74.22%	62.39%
Chunked FinBERT + BiLSTM	~53.6%	~53.5%	~99%	~69.5%
FinBERT - positive sentiment only subset	54.28%	–	–	–
LLaMA-based RF classifier	~51%	~54%	~59%	~57%
Ensemble (FinBERT + LLaMA agree Up)	**54.60%**	–	–	–

FinBERT variants generally underperform, with accuracy near or below baseline. Even when Up recall is high, precision is low (e.g., FinBERT test 1 with 100% recall but no discrimination). The chunking strategy and training more layers did not improve accuracy either, suggesting limited signal in the text or overfitting. **Performing the sentiment audit reveals ~59% positive, ~41% neutral, $<$ 0.1% negative predictions**. This confirms the "happy-talk" hypothesis, where earnings calls are dominated by upbeat or cautious-neutral phrasing, making binary drift prediction difficult unless sentiment shift is strong. Filtering to FinBERT's positive subset yields a modest lift (54.28%), but drift is still weak overall.

The LLaMA-predicted confidence labels, when used as features in classical classifiers such as logistic regression or random forest, also fail to beat baseline. Namely, LLaMA prompt engineering returned mostly "Up" confidence labels, so the classifier trained on counts struggles to generalize beyond default Up. Lastly, the ensemble strategy achieves the best accuracy of 54.60% (~1% above baseline) when both models agree on Up, suggesting complementary signals across approaches. One reason why this approach seems to work is because it reinforces only the strongest signals (when both models latch on to Up), and while it is more restrictive, this consensus appears to filter noise and yield modest accuracy gain.

4.2 Error Analysis & Discussion

4.2.1 Skewed predictions & overfitting

FinBERT's over-reliance on optimistic tone – especially in opening remarks – caused systematic bias: nearly all predictions are Up, irrespective of true direction. This was especially severe when only classifier layers were fine-tuned and indicates that training on a model pretrained for sentiment exaggerates existing tone biases.

4.2.2 Token-limited context

Using only the first chunk fails to capture the full narrative arc of earnings calls. Negative sentiment may appear later, and classifying only opening remarks limits signal capture. While chunked modeling theoretically addresses this, FinBERT representations remain insufficiently discriminative when the financial tone is consistently neutral/positive. This may be due to two reasons:

- Overfitting: Adding an LSTM over noisy CLS vectors from entire transcripts amplifies variance, and early-stopping could not avoid fitting to noise in minority Down cases.
- Signal dilution: Longer inputs dilute tone, and since executives often begin calls on upbeat notes – regardless of content – LSTMs may not be able to distinguish nuance across chunks given small training sets.

4.2.3 Weak signal in short horizon
The high baseline (~53.7%) and minimal lift in most experiments suggest that Close_5_dir is inherently noisy with weak linguistic proxies. Stronger drift prediction may require additional features – e.g., structured earnings surprises, guidance deviation, or investor sentiment responses, such as in Zhu et al. (2025).

4.2.4 Promise in ensemble alignment
When both FinBERT and LLaMA agree on Up, accuracy improves by ~1 percentage point. While modest, such lifts can yield meaningful financial value for long-only strategies. This suggests that agreement between different modalities (encoder + decoder) may filter for stronger signals. Our result aligns with Bai et al. (2023), where divergence from expected language reflects informative signals, but here agreement in Up may indicate consensus tone confidence.

4.2.5 Time-horizon behavior
We not only saw that Up accuracy was greater than respective baselines for all time horizons, but we also observed a trend that accuracy edge increases with longer prediction windows (Appendix A.3). This may reflect increased drift magnitude over time – making upward direction easier to forecast – though it could also simply be a more pronounced majority-class prevalence. Either way, our long-only strategy appears most justified in 1-5 day windows.

4.3 Implications for Deployment
A deployment that only utilizes Up prediction when both models agree yields better-than-baseline accuracy. Even though Down predictions are neglected (long-only strategy), this is acceptable for retail use where shorting is limited and risk asymmetric. Even a 1% edge in directional accuracy may translate to economically meaningful returns in algorithmic strategies, especially if applied consistently across calls.

That said, Down-side performance is poor, signaling the need for better architecture (e.g. hierarchical LLM fine-tuning, chunk-level attention), more data, proxy signals (such as HAID score or speaker tone features), or richer investor-response inputs (like interactive Q&A sections).

5 Conclusion

This study explored whether earnings call transcripts contain short-term predictive signals for post-earnings stock price movement. By isolating transcript language from structured data, we assessed both fine-tuned encoder-based models and decoder-style LLMs, including FinBERT and LLaMA-3.1. Despite strong biases toward the majority “Up” class, a combined strategy that requires agreement between both models yielded a modest but consistent accuracy improvement over baseline. These findings support the hypothesis that selective long-only predictions based on language agreement can yield a practical forecasting edge. Future work may expand this foundation using Q&A content, hierarchical attention mechanisms, and richer investor response features.

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Appendix:

A.1 Decoder-Based Predictions: Sample Outputs from LLaMA-3.1 Prompting

The table below illustrates collated outputs from the LLaMA-3.1 decoder model, where each row corresponds to a unique earnings call transcript. The first column indicates the transcript identifier, while the next three columns represent the three independent predictions generated via repeated prompting of the model. Each prediction is constrained to one of four possible responses: confident up, unconfident up, unconfident down, or confident down. These outputs form the basis for feature engineering in our downstream classification model, where the frequency of each prediction label (e.g., number of "confident up" responses) is used to infer directional drift.

	Transcript_Num	pred_1	pred_2	pred_3
0	0	unconfident down	unconfident up	unconfident up
1	1	confident up	unconfident up	confident up
2	2	confident up	confident up	unconfident down
3	3	confident up	confident up	confident up
4	4	unconfident down	confident up	confident up
...	...	...	...	...
17228	17228	unconfident down	confident up	confident up
17229	17229	unconfident down	unconfident down	unconfident down
17230	17230	unconfident up	unconfident down	confident down
17231	17231	confident up	unconfident up	confident down
17232	17232	unconfident up	confident up	confident up

Note: The "Up" predictions – both confident and unconfident – substantially outnumber the "Down" predictions. This imbalance contributes to a prediction skew toward the Up class in downstream models.

A.2 FinBERT vs. LLaMA: Directional Accuracy Comparison

This table compares directional prediction accuracy for both FinBERT and LLaMA-based models. The analysis separates predictions where each model independently predicted Up or Down, and evaluates their accuracy against the true Close_5_dir labels. The majority-class baseline for this task was 53.67%.

Model	Majority Baseline (%)	Accuracy of Up-Only Predictions	Accuracy of Down-Only Predictions
FinBERT	53.67%	54.28%	49.58%
LLaMA	53.67%	53.20%	48.93%

Note: Down-only performance is significantly lower, reinforcing the model's asymmetrical reliability.

A.3 Ensemble Strategy Accuracy by Time Horizon

This table reports accuracy of Up-only predictions made by the FinBERT + LLaMA ensemble model when both agree on the "Up" class, across various post-earnings time horizons. For each horizon, accuracy is compared to its respective majority-class baseline. The 5-day horizon (Close_5_dir) is the primary focus of the main paper.

Time Horizon (in Trading Days)	Majority Baseline (%)	Accuracy of Up-Only Predictions	Accuracy of Down-Only Predictions
5 days	53.67%	54.60%	50.96%
2 days	51.45%	53.03%	50.00%
1 day	50.76%	51.84%	50.00%
0 days (same day exit as entry)	50.65%	50.72%	48.80%

Note: Accuracy improvements persist across all horizons, with strongest gains observed in the longer window, potentially reflecting more stable drift effects.